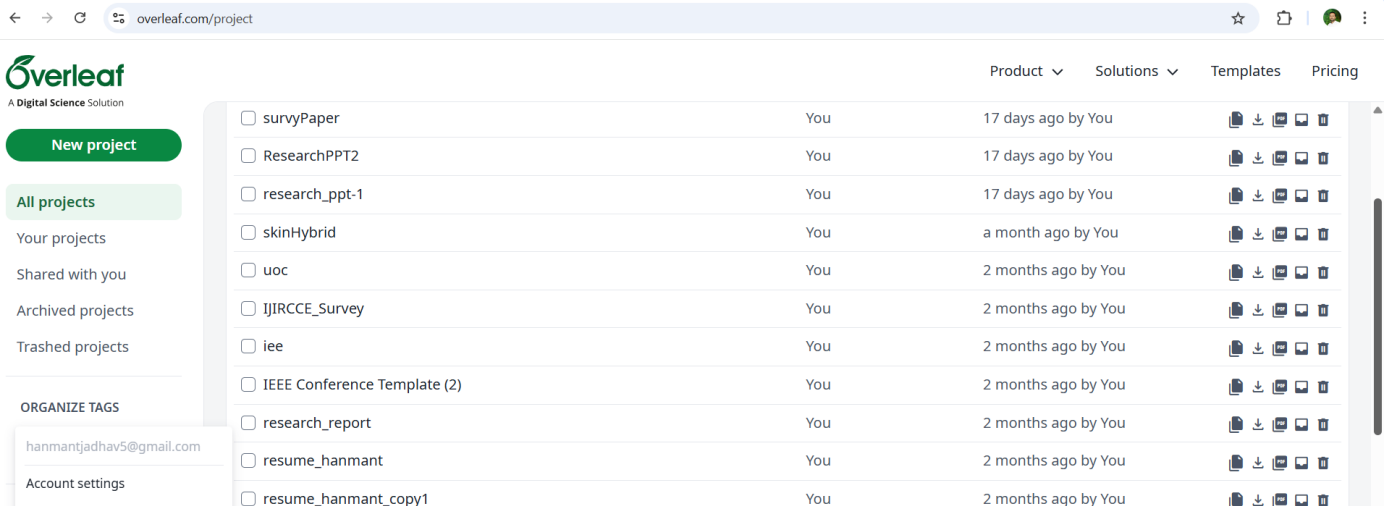


Assignment 3: Proposal Writing using Overleaf

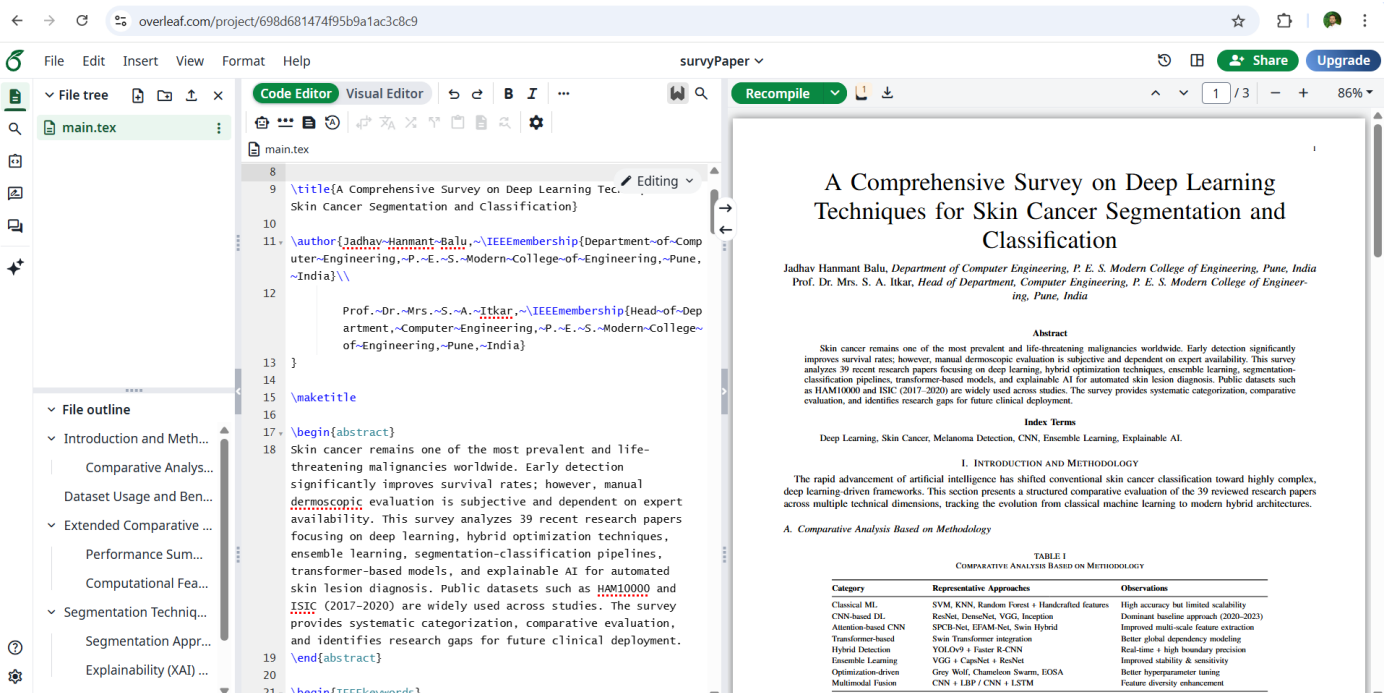
Step 1: Created Project in Overleaf



The screenshot shows the Overleaf dashboard. On the left, there's a sidebar with the Overleaf logo, a 'New project' button, and a list of 'All projects' including 'Your projects', 'Shared with you', 'Archived projects', and 'Trashed projects'. Below this is the 'ORGANIZE TAGS' section with a search bar and 'Account settings'. The main area displays a table of projects:

Project Name	Owner	Last Modified	Actions
survyPaper	You	17 days ago by You	Download, Upload, Share, Delete
ResearchPPT2	You	17 days ago by You	Download, Upload, Share, Delete
research_ppt-1	You	17 days ago by You	Download, Upload, Share, Delete
skinHybrid	You	a month ago by You	Download, Upload, Share, Delete
uoc	You	2 months ago by You	Download, Upload, Share, Delete
IJIRCEE_Survey	You	2 months ago by You	Download, Upload, Share, Delete
iee	You	2 months ago by You	Download, Upload, Share, Delete
IEEE Conference Template (2)	You	2 months ago by You	Download, Upload, Share, Delete
research_report	You	2 months ago by You	Download, Upload, Share, Delete
resume_hanmant	You	2 months ago by You	Download, Upload, Share, Delete
resume_hanmant_copy1	You	2 months ago by You	Download, Upload, Share, Delete

Step 2: Drafting the Research Paper in LaTeX



The screenshot shows the Overleaf Code Editor interface. The left sidebar contains a 'File tree' with 'main.tex' and a 'File outline' with sections like 'Introduction and Meth...', 'Dataset Usage and Ben...', 'Extended Comparative ...', 'Performance Sum...', 'Computational Fea...', 'Segmentation Techniq...', 'Segmentation Appr...', 'Explainability (XAI) ...', and 'Overall Comparative Te...'. The main editor shows the LaTeX code for 'main.tex' with line numbers 8 to 21. The right pane shows the compiled PDF of the document.

LaTeX Code (main.tex):

```
8 \title{A Comprehensive Survey on Deep Learning Tec.}
9 \title{Skin Cancer Segmentation and Classification}
10
11 \author{Jadhav-Hanmant-Balu,~\IEEEmembership{Department-of-Comp
12   uter-Engineering,~P.E.-S.-Modern-College-of-Engineering,~Pune,
13   ~India}}
14
15 Prof.~Dr.-Mrs.-S.-A.-Itkar,~\IEEEmembership{Head-of-Dep
16   artment,~Computer-Engineering,~P.E.-S.-Modern-College-
17   of-Engineering,~Pune,~India}
18 }
19 \maketitle
20 \begin{abstract}
21   Skin cancer remains one of the most prevalent and life-
22   threatening malignancies worldwide. Early detection
23   significantly improves survival rates; however, manual
24   dermoscopic evaluation is subjective and dependent on expert
25   availability. This survey analyzes 39 recent research papers
26   focusing on deep learning, hybrid optimization techniques,
27   ensemble learning, segmentation-classification pipelines,
28   transformer-based models, and explainable AI for automated
29   skin lesion diagnosis. Public datasets such as HAM10000 and
30   ISIC (2017-2020) are widely used across studies. The survey
31   provides systematic categorization, comparative evaluation,
32   and identifies research gaps for future clinical deployment.
33 \end{abstract}
34 \begin{IEEEkeywords}
```

PDF Content:

A Comprehensive Survey on Deep Learning Techniques for Skin Cancer Segmentation and Classification

Jadhav Hanmant Balu, *Department of Computer Engineering, P. E. S. Modern College of Engineering, Pune, India*
Prof. Dr. Mrs. S. A. Itkar, *Head of Department, Computer Engineering, P. E. S. Modern College of Engineering, Pune, India*

Abstract

Skin cancer remains one of the most prevalent and life-threatening malignancies worldwide. Early detection significantly improves survival rates; however, manual dermoscopic evaluation is subjective and dependent on expert availability. This survey analyzes 39 recent research papers focusing on deep learning, hybrid optimization techniques, ensemble learning, segmentation-classification pipelines, transformer-based models, and explainable AI for automated skin lesion diagnosis. Public datasets such as HAM10000 and ISIC (2017-2020) are widely used across studies. The survey provides systematic categorization, comparative evaluation, and identifies research gaps for future clinical deployment.

Index Terms

Deep Learning, Skin Cancer, Melanoma Detection, CNN, Ensemble Learning, Explainable AI.

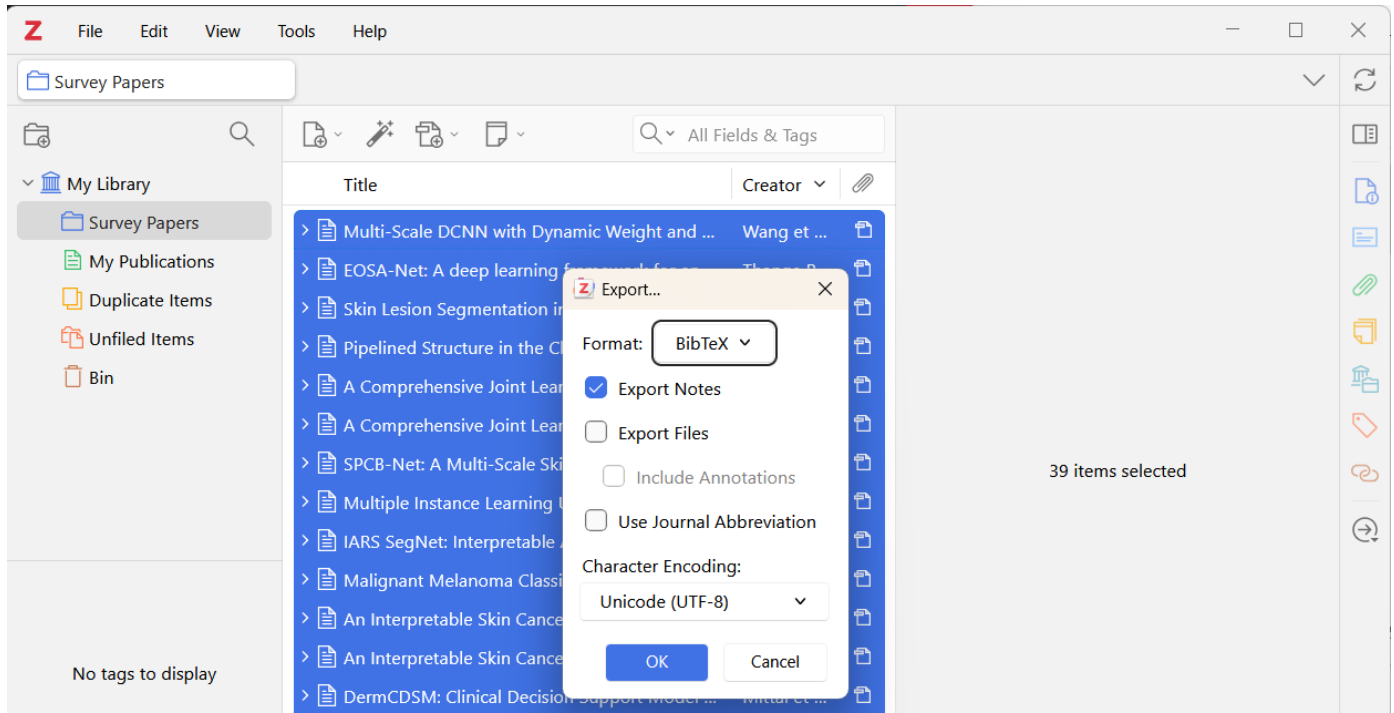
I. INTRODUCTION AND METHODOLOGY

The rapid advancement of artificial intelligence has shifted conventional skin cancer classification toward highly complex, deep learning-driven frameworks. This section presents a structured comparative evaluation of the 39 reviewed research papers across multiple technical dimensions, tracking the evolution from classical machine learning to modern hybrid architectures.

A. Comparative Analysis Based on Methodology

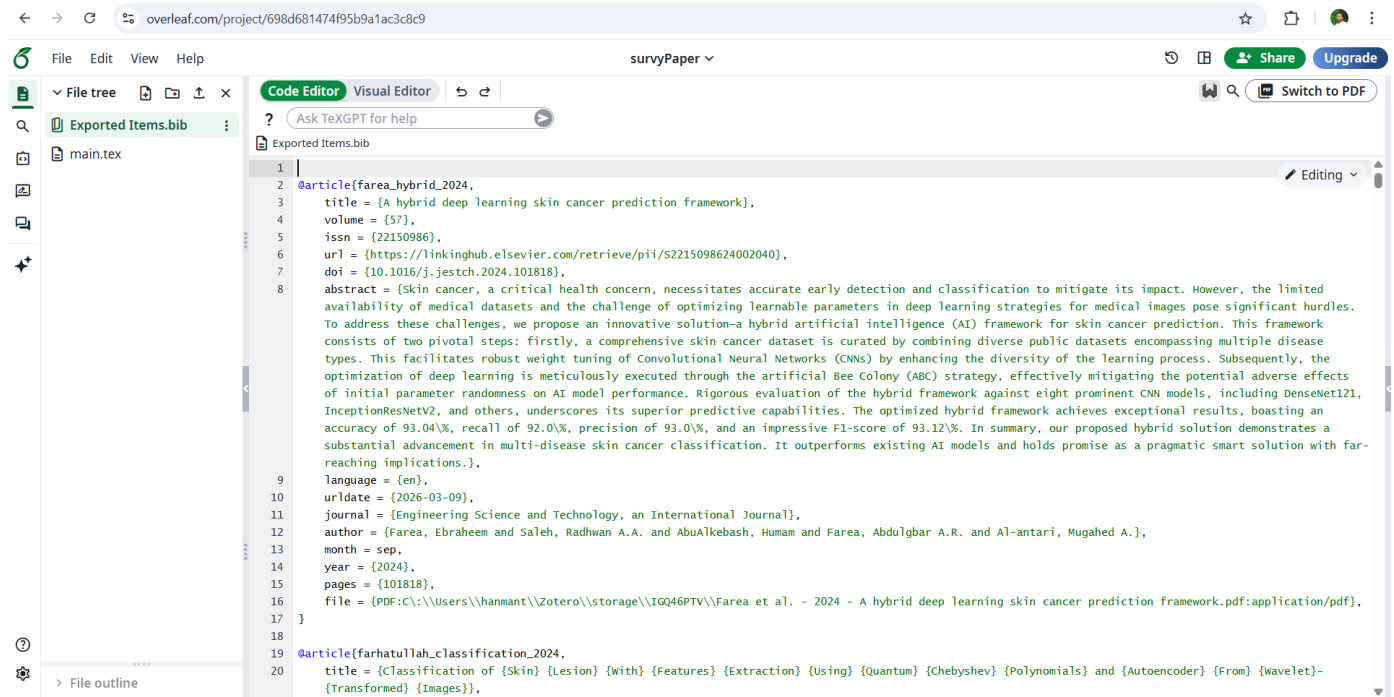
Category	Representative Approaches	Observations
Classical ML	SVM, KNN, Random Forest + Handcrafted features	High accuracy but limited scalability
CNN-based DL	ResNet, DenseNet, VGG, Inception	Dominant baseline approach (2015-2020)
Attention-based CNN	SPCB-Net, EFAM-Net, Swin Hybrid	Improved multi-scale feature extraction
Transformer-based	Swin Transformer integration	Better global dependency modeling
Hybrid Detection	YOLOv9 + Faster R-CNN	Real-time + high boundary precision
Ensemble Learning	VGG + CapsNet + ResNet	Improved stability & sensitivity
Optimization-driven	Grey Wolf, Chameleon Swarms, ESOA	Better hyperparameter tuning
Multimodal Fusion	CNN + LBP / CNN + LSTM	Feature diversity enhancement

Step 3: Exporting the Zotero Library to BibTeX



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4 |   volume = {57},
5 |   issn = {22150986},
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7 |   doi = {10.1016/j.jestch.2024.101818},
8 |   abstract = {Skin cancer, a critical health concern, necessitates accurate early detection and classification to mitigate its impact. However, the limited availability
9 |   language = {en},
10 |   urldate = {2026-03-09},
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12 |   author = {Farea, Ebraheem and Saleh, Radhwan A.A. and AbuAlkebash, Humam and Farea, Abdulgbar A.R. and Al-antari, Mugahed A.},
13 |   month = sep,
14 |   year = {2024},
15 |   pages = {101818},
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17 | }
18 |
19 | @article{farhatullah_classification_2024,
20 |   title = {Classification of {Skin} {Lesion} {With} {Features} {Extraction} {Using} {Quantum} {Chebyshev} {Polynomials} and {Autoencoder} {From} {Wavelet}-{Transformed}
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24 |   url = {https://ieeexplore.ieee.org/document/10757392/},
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26 |   abstract = {Skin, a vital organ acting as a protective barrier to the external environment, plays a pivotal role in overall human health. Early detection of skin dise
27 |   language = {en},
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29 |   journal = {IEEE Access},
30 |   author = {{Farhatullah} and Chen, Xin and Zeng, Deze and Xu, Jiafeng and Nawaz, Rab and Ullah, Rahmat},
31 |   year = {2024},
32 |   pages = {193923--193936},
33 |   file = {PDF:C:\Users\hanmant\Zotero\storage\H9828PP6\Farhatullah et al. - 2024 - Classification of Skin Lesion With Features Extraction Using Quantum Chebyshev
34 | }
35 |
36 | @article{kanthik hybrid 2024
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Step 4: Integrating the Bibliography into Overleaf



Step 5: Citation and References

1. Introduction

Skin cancer is a prevalent public health concern that affects millions of individuals globally, regardless of age and skin type. Melanoma, also known as malignant melanoma, is one of the most dangerous forms of skin cancer [1]. It is primarily caused by long-term exposure to UV radiation, which can lead to mutations in the cells that produce melanin, called melanocytes. These mutations can cause cells to produce excessive melanin, resulting in dark moles on the skin. Over time, moles or lesions may progress and transform into cancerous tumors that can spread and wreak havoc in other body parts. If left unchecked, they can become uncontrollable and seriously threaten an individual's health and well-being. The incidence rate is currently around 23 per 100,000 in the U.S. However, early detection and proper treatment of melanoma can significantly reduce the morbidity and mortality associated with the disease. Numerous studies have demonstrated that if melanoma is detected in its early stages, the prognosis is much more

- [1] L. Riaz *et al.*, 'A Comprehensive Joint Learning System to Detect Skin Cancer', *IEEE Access*, vol. 11, pp. 79434–79444, 2023, doi: [10.1109/ACCESS.2023.3297644](https://doi.org/10.1109/ACCESS.2023.3297644).
- [2] L. Riaz *et al.*, 'A Comprehensive Joint Learning System to Detect Skin Cancer', *IEEE Access*, vol. 11, pp. 79434–79444, 2023, doi: [10.1109/ACCESS.2023.3297644](https://doi.org/10.1109/ACCESS.2023.3297644).
- [3] R. Karthik, R. Menaka, S. Atre, J. Cho, and S. Veerappampalayam Easwaramoorthy, 'A Hybrid Deep Learning Approach for Skin Cancer Classification Using Swin Transformer and Dense Group Shuffle Non-Local Attention Network', *IEEE Access*, vol. 12, pp. 158040–158051, 2024, doi: [10.1109/ACCESS.2024.3485507](https://doi.org/10.1109/ACCESS.2024.3485507).
- [4] R. Karthik, R. Menaka, S. Atre, J. Cho, and S. Veerappampalayam Easwaramoorthy, 'A Hybrid Deep Learning Approach for Skin Cancer Classification Using Swin Transformer and Dense Group Shuffle Non-Local Attention Network', *IEEE Access*, vol. 12, pp. 158040–158051, 2024, doi: [10.1109/ACCESS.2024.3485507](https://doi.org/10.1109/ACCESS.2024.3485507).
- [5] E. Farea, R. A. A. Saleh, H. AbuAlkebash, A. A. R. Farea, and M. A. Al-antari, 'A hybrid deep learning skin cancer prediction framework', *Engineering Science and Technology, an International Journal*, vol. 57, p. 101818, Sep. 2024, doi: [10.1016/j.jestch.2024.101818](https://doi.org/10.1016/j.jestch.2024.101818).
- [6] M. I. Marie, M. S. Elredeney, and A. Essayed Yakoub, 'A Hybrid Model for Early Melanoma Detection: Integrating YOLOv9 and Faster R-CNN for Enhanced Diagnostic Accuracy', *IEEE Access*, vol. 13, pp. 120327–120344, 2025, doi: [10.1109/ACCESS.2025.3587625](https://doi.org/10.1109/ACCESS.2025.3587625).
- [7] M. I. Marie, M. S. Elredeney, and A. Essayed Yakoub, 'A Hybrid Model for Early Melanoma Detection: Integrating YOLOv9 and Faster R-CNN for Enhanced Diagnostic Accuracy', *IEEE Access*, vol. 13, pp. 120327–120344, 2025, doi: [10.1109/ACCESS.2025.3587625](https://doi.org/10.1109/ACCESS.2025.3587625).